TOPIC 2

APPROACH IN PROBLEM SOLVING

COMPUTER SCIENCE II| SC025

Prepared by: Ms Jasini



Learning Outcome

At the end of the lesson, students should be able to:

a) Describe steps in problem solving: **problem analysis**, **design a solution**, **implementation**, **testing** and **documentation**.

b) Identify **input**, **process**, and **output** based on the given problem statement.

Introduction

Solving problems is the **core** task of **computer programming**.

Below are the **five** (5) **approaches in problem solving**:

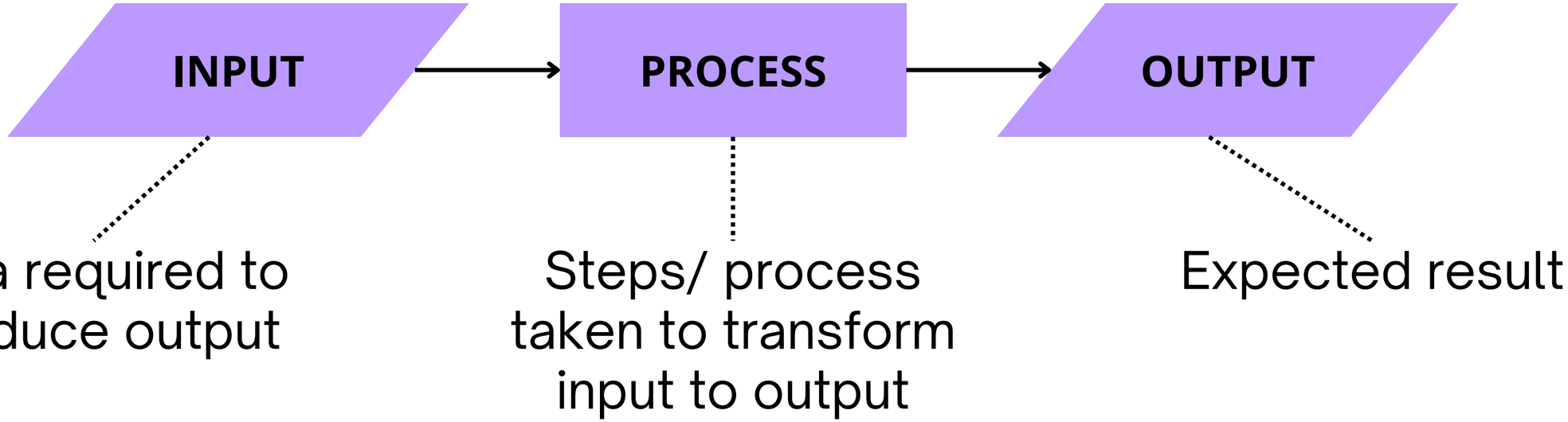


Start by identifying and understanding the problem. Only then can we develop and test solutions accordingly.

Step 1: Problem Analysis

Definition:

Problem analysis is the process of **identifying** **input**, **process**, and **output** for a given problem statement.



Step 1: Problem Analysis

Given a problem statement, the **input** is the **data** needed to obtain the **required output**.

Understanding Input

- There can be **more than one input**.
- Depending on the problem, there also may be **no input**.
- A **constant** (known value) **cannot** be an **input**.
- Input must be **entered by the user**.
- Keyword indicators in statement: **input, read, get, from...**

Step 1: Problem Analysis

Given a problem statement, the **process** is the **transformtion** of **input** into **output**.

Understanding Process

- Any activity that changes the input into output.
- Three main control structures involved:
  - Sequence
  - Selection
  - Repetition
- Keyword indicators in statement: **calculate, compute, determine, check, find, compare, while, if, until, loop...**

Step 1: Problem Analysis

Given a problem statement, the **output** is the **expected result** from the problem statement.

Understanding Output

- There **MUST** be **at least one output**.
- There are two types of output:
  - **Value output**
    - e.g. salary, total price (RM), area, average etc...
  - **Message output**
    - e.g. “hello world”, “Even” or “Odd” etc...
- Keyword indicators in statement: **output, display, show, print..**

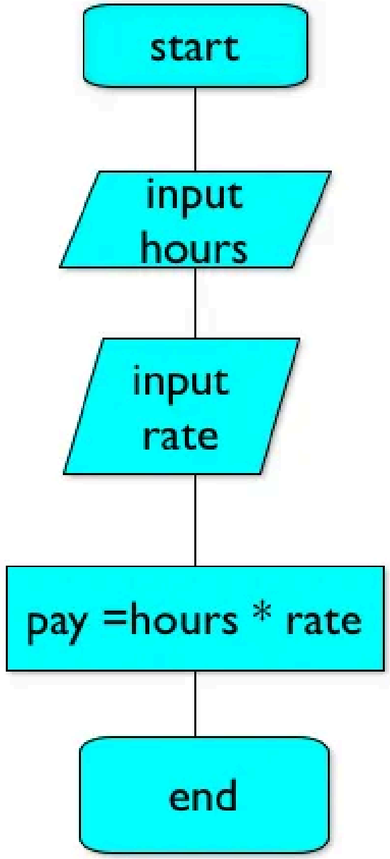
Step 2: Design a Solution

- Designing a solution involves creating an **algorithm**.
- An **algorithm** is a precise **sequence of well-defined steps** to **solve a problem**.
- There are **two methods** used to **demonstrate** the **logical steps** in an **algorithm**:
  - ***Flowchart*** - *graphical representation*
  - ***Pseudocode*** - *English-like representation*
- This will be the focus in **Chapter 3**.

Step 2: Design a Solution

Example:

Flowchart



Pseudocode

```
BEGIN
    input hours
    input rate
    pay = hours * rate
    print pay
END
```

Step 3: Implementation

- **Implementation** is the **application** of the **algorithm** by **writing** a **computer program** using a **programming language**.

Additional Info

- During this course, we will be using **Java** as our **main programming language**.
- To implement computer programs we will **write our code** in a text editor or **IDE** (Integrated Development Environment) such as BlueJ, jGRASP, Eclipse or others. [We will mainly use **BlueJ** in this course]
- We will explore more on this during **Chapter 4**.

Step 4: Testing

- **Testing** is the process of making sure our **programs work correctly** and does what it is supposed to.
- **Functions**
  - **Identify errors** - Find mistakes and bugs in the program
  - **Verify outputs** - Ensure program produces correct results
  - **Fix issues** - Fix mistakes in the program (Debugging)
  - **Confirm requirements** - Make sure the program meets the needs and expectations of the user

Step 5: Documentation

- **Documentation** contains **description** of a **program** that **helps programmers understand**, **edit**, or **maintain** the program in the future.
- There are two (2) ways of program documentation:
  - **Comments**
    - Different programming languages have different ways to write comments in code.
    - It is considered good programming habit to include comments inside your code that explains what each part does.
  - **Text file**
    - A text file that contains the programs' requirements, user manual, and operational manual.

Practice Questions

Identify the steps in problem solving for the following statements.

Ahmad starts writing code for a computer program that prints out students' results.	
Identifying the processes involved in creating a certain program.	
Wallace writes comments in his code so he does not forget their functions later.	
Drawing a flowchart to represent the steps needed in an area calculator program.	
Lakshmi is correcting her code after realizing the output was wrong.	

FAQs

1. Describe the five (5) steps in problem solving.
2. Identify the step of problem solving based on given statements.
3. Identify input, process, and output for given problem statements.